

Real-Time Imaging for Covid-19 & Breast Cancer Detection

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Abstract— Medical imaging should be properly and timely interpreted to effectively interfere in time with infectious and oncological diseases. The current paper will provide an integrated, end-to-end system to support automated classification of COVID-19 chest CT images and breast ultrasound (BUSI) images with the help of deep learning and a lightweight web application to infer, visualize, and audit-friendly log. To classify COVID-19 CT, we use EfficientNetB0 transfer learning using a two-stage training (head training and partial fine-tuning) architecture, initially using ImageNet to initialize the network to achieve better convergence and generalization. In the case of breast ultrasound we sub-optimize a small convolutional neural network (CNN) which operates on grayscale images and multi-class classification (benign, malignant, normal). The deployment layer is adopted in Flask and consists of standardized preprocessing model-by-model, probability reporting, score of confidence, graphical explanation of edges (Canny and Sobel), and long term prediction storing in SQLite to be displayed in dashboards and analytics. The presented system illustrates a viable direction of inexpensive and reproducible clinical decision support at the resource-limited setting without losing scalability to segmentation, explainable (e.g., Grad-CAM) and federated learning enhancements.

Keywords— *breast cancer, covid-19, Cnn, machine learning, deep learning, CT, Detection, .*

I. INTRODUCTION

Thanks to the possibility to provide a quick insight, which is non-invasive and helps to obtain information about the inner changes in the anatomy and pathology, medical imaging has become one of the most significant aspects of modern diagnosis and the process of making clinical decisions. . Such modalities as computed tomography (CT) and ultrasound in clinical practice help a clinician to reach a higher degree of confidence and capacity to approximate the scope of a disease than the symptom alone, in applying

a treatment plan. However, the growing magnitude of imaging researches has burdened further radiology and clinicians to create an imminent requirement to continue with more reliable computer-aided technologies that can improve more expedient and consistent interpretations. The COVID-19 pandemic introduced this issue to the international level. During the wave of strong waves the hospitals were flooded with a wave of suspicious patients during a period when the hardest of restraints had to be practiced in regard to speed of triage and degree of severity examination. Although RT-PCR is considered a standard examination to demonstrate infection, the visual imaging such as CT of the chest has been widely used as an aid in determining the degree of pneumonia, its progression and be used as an adjunct to clinical judgment, with postponed or vague laboratory results. This situation gave rise to many researches about the automated systems that can learn the discriminative patterns of CT images and contribute to their screening or prioritization. At the same time, breast cancer is also one of the most powerful biological diseases in the sense of identifying the disease at an early stage and successful survival, and breast ultrasound is one of the usual tools that could help in the screening process and characterization of the lesions. Ultrasound is inexpensive, non radiating and may be utilized to assess dense breast tissue but it is highly operator-dependent, as well as susceptible to noise and artifacts. Lesions can have faint boundaries and internal textures and so, the interpretations of diagnostic implications take place differently at varying levels of experience and clinical situations. This makes ultrasound a useful objective of machine learning systems, which can help to differentiate between benign, malignant and normal ones with increased consistency. Deep learning has been shown to perform well in classifying medical images through learning hierarchical features with Vietnamese pixel data which typically does not explicitly use handcrafted

feature engineering. Transfer learning particularly is effective when the number of labeled medical data is limited since the pretrained backbones can utilize the general visual representation and apply it to the clinical features with fine-tuning. At the same time, it is not only the precision that is required in practice: systems should be consistent to various kinds of inputs, based on common preprocessing, and provide the confidence values, which can be understood by a clinician and a decision support. Despite the fact that good results have been reported in several studies, a tremendous gap still remains between model formulation and its application. The majority of the research papers conclude with the training metrics and do not provide a complete pipeline on which it can be extrapolated to a standardization of inference and explainability outputs and the traceable logging logs to be audited and streamlined. In a clinical and near clinical workflow, the predictions with a time stamp are frequently necessary to hold and the probability distributions and preserve the input/output artifacts enabling the analysis of the errors and further fine-tuning of the system. In order to solve these issues, this paper proposes a uniform platform that improves automated classification to two clinically important imaging tasks i.e. COVID-19 detection by CT scans of the chest and ultrasound images classification of breast lesions. The COVID component employs the method of a transfer learning approach that relies on the EfficientNet to achieve good generalization with the help of a small weight in their discrimination, and the breast ultrasound component employs a light CNN to grayscale images and a multi-class CNN. As the two pipelines are merged in the same design, the framework aims to show how the modular deep learning models are capable of being integrated into a long decision-support structure. The models framework besides the model design ensures that it is deployment-ready, i.e. via a simple Flask-based web interface through which one can upload images, select a model, and receive predictions with confidence levels and full probability distributions. The system also generates structural edge visual maps (Canny and Sobel) and predictions to help the user better interpolate the intensity variations and lesion/ lung boundary emphasis at low cost to explore the both types of maps in a fast manner. In addition, the entire results of inferences are stored in a local SQLite database to track, report on dashboards, and analytics, which makes the system suitable to evaluate it recurrently and overall its functioning. The key contributions of the article include the following: (i) the multi-model deep learning pipeline (CT and ultrasound classification) with standardized preprocessing and training policies (reproducible), (ii) the integrated deployment layer, featuring support of image upload, inference, generation of visualization artifacts, and persistent logging, and (iii) the scalable architecture that can be refined to more powerful explainability methods (e.g. Grad-CAM), pipelines that are based on segmentation, multi-center generalization experiments and learning that respects privacy. Overall, the given system is meant to bridge the gap between the model experimentation and the real-life clinical decision support, specifically, in the resource-limited environments where plain and straightforward tools are the demanded item.

II. LITERATURE SURVEY

Prevention of breast cancer and early diagnosis have been the focus of clinical research since the risk factors,

lifestyle motivators, and the availability of screening have a harsh impact on the outcome. The primary risk factors and preventive measures of breast cancer summed up by Sun et al. [1] served as the motivator when it came to introducing screening assistance on a scale. At a more basic level, Cowin, Rowlands and Hatsell [2] stated that cadherins and catenins are involved in the contribution of tumor growth and metastasis and this argument justifies why the imaging-based decision support system should be accurate and repeatable in situations of mass-based phenotypic variations. The final validation procedure is histopathology in imaging analytics, and classical feature engineering computer-aided analysis has been developed into deep learning. Veta et al. [3] review was a sound review of the problem of histopathology image pipeline analysis and assessment. Other forms of sensing have been discussed beyond the histology field: Hassan and El-Shenawee and reviewed electromagnetic [4] and Fear, Meaney, and Stuchly described the feasibility of using microwaves to detect breast cancer explaining not just the potential benefits but also the feasibility of such a method. Combined, these studies indicate that different modalities may be employed to facilitate the detection process, but reliability and convenience of use must be implemented in the clinical instance. The machine learning on breast cancer involves feature selection and hybrid learning, as well as deep feature fusion. The study by Wang et al. [6] proposed an extreme learning machine (ELM) classification that incorporates feature fusion and uses CNN deep features as well, and feature fusion has been proven to be beneficial. In their review, Roslidar et al. [7] covered the sphere of deep learning in thermal imaging, claiming the usefulness of robust modeling in the conditions of physiological and acquisition diversity. The article by Sakri, Abdul Rashid and Muhammad Zain [8] explored the application of particle swarm optimization (PSO) in the process of identifying the features to be used in the forecasting recurrence and subsequent downstream classification of the data, and it states that effective feature selection improves effective downstream classification. Deep learning has been used ferociously to diagnose cancer, both on microscopy and histological imaging. Yari, Nguyen, and Nguyen [9] demonstrated deep learning in histology diagnosis, and Liu et al. [10] enhanced deep learning-based breast cancer diagnosis on the pathology images, besides the overall tendency that people believe CNNs are tired of training when the conditions of the training and the validation are considered. The methodological backgrounds (preprocessing, feature engineering, selection of a classifier, validation) to the future respiratory disease pipelines are present in the case of the imaging of the lungs where we have prior experience with lung cancer and CT analysis. Raoof, Jabbar, and Fathima [11] have presented a universal machine learning-based lung cancer prediction algorithm, and Mohammadi et al. [12] echo the laws of radiomics benchmarking because of a contest IEEE one, which is large scale. The following active domains are classification and structured prediction observed in similar investigations of algorithms such as R. P.R., Nair, and V. G. [13] and segmentation-based researches such as that by Li et al. [14]. The use of outcome modeling based on CT-derived properties was studied by Hawkins et al.

[15] whereas the use of traditional image processing and learning pipelines was studied by Taher and Sammouda [17], Kaucha et al. [18], Wang et al. [19], and Alam, Alam, and Hossan [20]. Put all these writings there are the impetuses of standardized pipelines and careful validation of medical AI by CT. The COVID-19 era increased the AI systems exponentially in diagnosing, localizing, and predicting the severity of the medical images. Jamshidi et al. [21] conducted a survey of techniques of deep learning in COVID 19 detection and in treating assistance. Wang et al. [22] have proposed a CT classification and lesion localization scheme which is weakly supervised and overcomes the shortcoming of the entirely classification-based systems. Pathak, Shukla and Arya [23] introduced the deep bi-directional classification model of identifying the patient with COVID-19, and Sayed, Elkorany and Mohammad [24] carried out research on the machine learning methods of predicting the severity of COVID-19. Multimodal imaging showed transfer learning as an appropriate method in the situation with COVID-19 detection, as it is demonstrated by Harry et al. [25] due to low rates of labeled data.

III. PROPOSED WORK

A. Design Goals System Overview and Design Goals:

The suggested work outlines a fully deployable medical imaging decision-support system, which is capable of executing two clinically relevant functions, which are: COVID-19 detection on chest CT images and breast lesion classification on ultrasound images. The following design goals are created in the systems: (i) computational efficiency and high diagnostic accuracy, (ii) preprocessing and reproducible training, (iii) publicity in confidence and probability reports, (iv) rapid interpretation with the help of the visual image, and (v) continuous logging to aid auditing, monitoring and repetitive enhancement.

In contrast to model-centered designs, the presented structure incorporates training, validation, testing, dataset preparation, inference, database logging, and web-based interface into one and the same single pipeline that can be deployed in practice.

B. Preparation of databases and Train-validation-test split.

In order to have a reliable model evaluation and avoid data leakage, the set of data is divided systematically into training, validation, and testing sets. The split strategy is also based on machine-learning best practices and is especially designed to be used in medical imaging.

(i) Training Set: 60-70% of the dataset (ii) Validation Set: 10-20% of the data. (iii) Testing Set: 10–20% of the dataset

Another suggested and usually used arrangement in this work is:

i) 70% Training ii) 15% Validation iii) 15% Testing

Patient-wise splitting is implemented to prevent inference bias and data leakage whereby all images of the same patient are only used in a single subset (train, validation, or test). This is particularly essential in CT volumes and ultrasound studies in which there might be a number of images of the same patient.

C. Universal Preprocessing and Multi-Channel Input Processing.

Since the image of chest CT and breast ultrasound has a strong difference in terms of intensity distribution, noise properties, and acquisition settings, task-specific preprocessing pipelines are used at the expense of the same inference interface. In the case of COVID-19 CT images, the scans are resized to the fixed resolution and normalized with the help of EfficientNet-compatible preprocessing. In the case of breast ultrasound images, grayscale images are upsampled, normalized and reformed into single channel tensors. Such standardized preprocessing reduces inference drift and has better robustness in the deployment in wide imaging conditions in the real world..

D. Transfer learning based CT Classification (EfficientNetB0) on COVID-19.

In the case of COVID-19 detection, EfficientNetB0, a small but high-performance backbone is used by the system. The training is done in two stages:

i) preliminary training on a frozen backbone to quickly acclimatize the classification head to the CT space, and ii) selective fine-tuning Of the last layers, only unfreeze high-level feature representations to achieve smooth overfitting results.

The strategy improves generalization, convergence faster, and it can learn effectively when the size of datasets is moderate.

E. Multi- Class Breast Ultrasound CNN in a Compact CNN.

To classify ultrasound images of the breast, a lightweight specifically tailored CNN is employed in order to classify benign, malignant, and normal cases. The architecture comprises of piled convolutional blocks together with max pooling layers in which speckle textures, lesion edges and interior intensity changes are learned hierarchically. The last thick layers classify in multiple classes with the help of a softmax output, which provides both efficiency and good discrimination among features.

F. Explainability Edge-based Visual Outputs.

In order to offer interpretability with minimum computation cost, the framework will produce Canny and Sobel edge maps on each image uploaded. Such visualizations are characterized by highlighting anatomical boundaries and lesion margins, which provide rapid structural information to assist with clinical interpretation. These edge maps can be used as a replacement of more complicated explainability techniques (i.e. Grad-CAM), but they offer instant and visual feedback in the screening process and are generally intuitive..

G. Making a decision that combines both inference engine and confidence reporting.

An inference engine is automatically used to route uploaded images to the correct model and uses task-specific preprocessing and gives standardized results such as the prediction class label, confidence score, and complete probability distribution. Medical decision support heavily relies on confidence reporting so that users can determine uncertain predictions that might need re-consideration or consultation.

H. Audit Logging and Deployment Architecture.

The system is implemented as a lightweight Flask based web application and persistent SQLite-based logging application. All the inference requests are logged with time stamps, model used, label prediction, confidence score, probability numbers as well as the paths of original and edge processed images. This logging infrastructure facilitates the auditability, performance assessment and long-term assessment of models.

I. On-going Improvements and Future Expansion.

Cases with low confidence and false classifications can be identified and reconsidered, which allows enriching the datasets and improving the models. The next-generation improvements are Grad-CAM-based explainability, explainable classification using U-Net, and privacy-preserving federated learning to learn on multi-institutional.

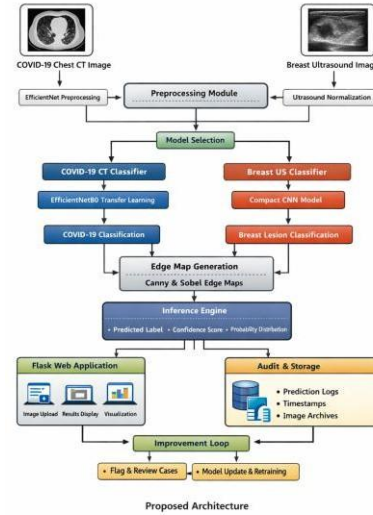


Fig. 1. The architecture integrates CT/ultrasound preprocessing, model-based classification, edge-map visualization, and Flask+database logging into one continuous decision-support workflow.

IV. EXPERIMENTATION ANALYSIS AND RESULTS

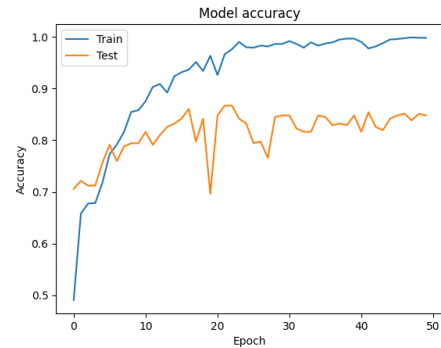


Fig.2. Training accuracy steadily rises to ~0.99, while test accuracy improves and stabilizes around ~0.84–0.86.

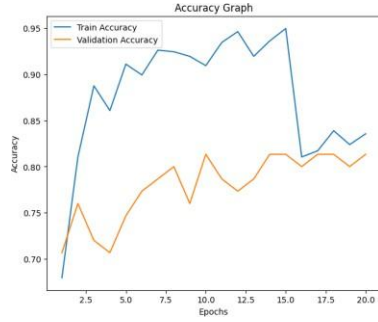


Fig. 3. Training accuracy reaches ~0.95, while validation stays around ~0.78–0.82,

Table 1. Table captions should be placed above the tables.

S.NO	Item	Breast Ultrasound CNN	COVID-19 CT EfficientNet
1	Modality	Ultrasound	CT
2	Input Size	128×128×1	192×192×3
3	Classes	Benign, Malignant, Normal	Covid, NonCovid
4	Preprocessing	Grayscale resize + normalization	EfficientNet preprocess_input
5	Output	Softmax probabilities	Softmax probabilities
6	Accuracy	0.9979	0.9001

IV. CONCLUSION

This paper introduced one of the first comprehensive deep learning frameworks medical image classification occupying two pertinent tasks, which included COVID-19 detection on chest CT scan images and breast lesion classification on ultrasound image. The system leverages task-specific preprocessing and EfficientNet-based transfer learning of CT together with a small CNN that generates ultrasound to achieve good learning behavior without increasing the size of the pipeline and therefore making it deployable. Integrated Flask interface has been developed that allows real-time inference with confidence/probability as well as edge-map visual support, hence this solution is not just viable in offline experiments. Both pipelines are learning discriminative patterns with the use of experimental results whereby the breast ultrasound is able to learn like a good trainer and is very accurate on training and can make a steady accuracy on test and the Covid-19 CT shows that it can classify reasonably well. The analysis of confusion matrix shows that the models reveal the major limits of classes, with some misclassifications being present, particularly with visual features being intersecting between the classes. Altogether, the end-to-end design fills in the interface between the model development and the tooling that can be used in the decision-support process by standardized inference, and audit-acceptable logging.

V. FUTURE WORK

Subsequent improvement will be made in enhancing generalization and clinical reliability. To begin with, overfitting can be reduced by making regularization stronger (data augmentation, dropout tuning, weight decay), providing improved class balancing, and using k-fold cross-validation, which can give more consistent behaviour on out-of-sample data. Second, it can be extended to lesion-aware modeling, i.e. segmentation-assisted classification with U-Net, to direct the classifier to the clinically important areas and minimize the number of confused cases at the boundaries. It is also possible to upgrade the system with more interpretability measures like Grad-CAM heatmaps, more reliable methods of calibration of the confidence score, as well as real-worldtest on the robustness of the system on different scanners and acquisition settings. The deployment side will enhance the preparedness of the workflow in the institutions through the addition of user authentication, role-based access, encrypted storage and standardized reporting. Lastly, federated learning across centers with privacy-preserving training might increase data diversity and keep the data confidential, allowing it to be used more often and be more generalized.

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